Vendor Reliability Intelligence Platform

Database Design Report – Milestone 1

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Submission Date: *10 July 2026*

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1. Introduction

The Vendor Reliability Intelligence Platform is a web application designed to help organizations manage vendors, procurement processes, contracts, and vendor performance. The database is an important part of the system because it securely stores and organizes all the information required for the application.

The database is designed using PostgreSQL and follows a relational database model. It maintains relationships between users, vendors, procurement requests, purchase orders, contracts, reports, and notifications.

2. Database Requirements

The database should support the following functionalities:

- *Store user information and authentication details.*
- *Manage different user roles such as Administrator, Procurement Manager, Vendor, Finance Officer, and Auditor.*
- *Maintain vendor information and contact details.*
- *Store procurement requests and purchase orders.*
- *Track vendor performance and reliability scores.*
- *Manage contracts and compliance records.*
- *Store notifications sent to users.*
- *Generate procurement and vendor reports.*

3. Database Tables

Table Name	***Purpose***
Roles	*Stores different user roles*
Users	*Stores user account information*

Vendors	*Stores vendor details*
Procurement Requests	*Stores procurement requests*
Purchase Orders	*Stores purchase order information*
Vendor Performance	*Stores vendor performance ratings*
Contracts	*Stores contract information*
Notifications	*Stores user notifications*
Reports	*Stores generated reports*

4. Table Descriptions

4.1 Roles Table

Purpose: *Stores all user roles available in the system.*

Column	***Description***
role_id	*Primary Key*
role_name	*Name of the role*

4.2 Users Table

Purpose: *Stores registered users.*

Column	***Description***
user_id	*Primary Key*
full_name	*User's full name*
email	*User email*
password	*Encrypted password*
phone	*Phone number*
role_id	*Foreign Key*

created_at	Registration date

4.3 Vendors Table

Purpose: Stores vendor details.

Column	Description
vendor_id	Primary Key
vendor_name	Vendor name
category	Vendor category
contact_person	Contact person
email	Email
phone	Phone number
address	Vendor address
status	Vendor approval status

4.4 Procurement Requests Table

Purpose: Stores procurement requests.

Column	***Description***
procurement_id	*Primary Key*
title	*Procurement title*
description	*Procurement description*
vendor_id	*Foreign Key*
status	*Current status*
created_date	*Request date*

4.5 Purchase Orders Table

Purpose: *Stores purchase order details.*

Column	***Description***
order_id	*Primary Key*
vendor_id	*Foreign Key*

amount	*Order amount*
order_date	*Date of order*
delivery_date	*Delivery date*
status	*Order status*

4.6 Vendor Performance Table

Purpose: *Stores vendor ratings.*

Column	***Description***
performance_id	*Primary Key*
vendor_id	*Foreign Key*
delivery_rating	*Delivery performance*
quality_rating	*Product quality*
communication_rating	*Communication*

overall_score	*Final score*

4.7 Contracts Table

Purpose: *Stores contract information.*

Column	***Description***
contract_id	*Primary Key*
vendor_id	*Foreign Key*
start_date	*Contract start*
end_date	*Contract end*
status	*Contract status*

4.8 Notifications Table

Purpose: *Stores system notifications.*

Column	***Description***
notification_id	*Primary Key*

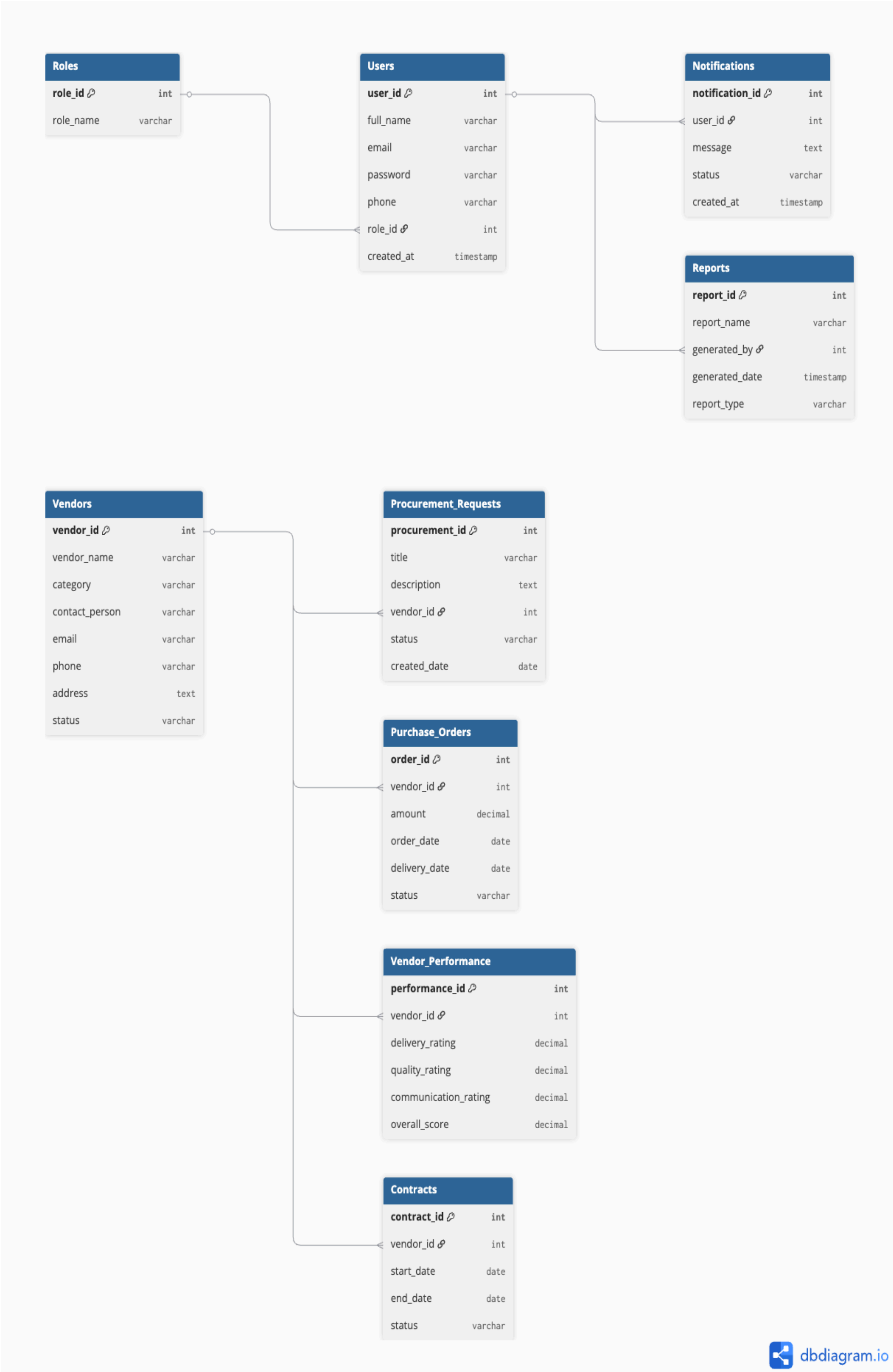
user_id	*Foreign Key*
message	*Notification message*
status	*Read/Unread*
created_at	*Date and time*

4.9 Reports Table

Purpose: *Stores generated reports.*

Column	***Description***
report_id	*Primary Key*
report_name	*Report title*
generated_by	*User who generated the report*
generated_date	*Date of generation*
report_type	*Report category*

5. ER diagram


Roles
role_id int
role_name varchar
Users
user_id int
full_name varchar
email varchar
password varchar
phone varchar
role_id int
created_at timestamp
Notifications
notification_id int
user_id int
message text
status varchar
created_at timestamp
Reports
report_id int
report_name varchar
generated_by int
generated_date timestamp
report_type varchar
Vendors
vendor_id int
vendor_name varchar
category varchar
contact_person varchar
email varchar
phone varchar
address text
status varchar
Procurement_Requests
procurement_id int
title varchar
description text
vendor_id int
status varchar
created_date date
Purchase_Orders
order_id int
vendor_id int
amount decimal
order_date date
delivery_date date
status varchar
Vendor_Performance
performance_id int
vendor_id int
delivery_rating decimal
quality_rating decimal
communication_rating decimal
overall_score decimal
Contracts
contract_id int
vendor_id int
start_date date
end_date date
status varchar
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6. Database Relationships

- One Role can be assigned to many Users.
- One Vendor can have many Procurement Requests.
- One Vendor can have many Purchase Orders.
- One Vendor can have many Performance Records.
- One Vendor can have many Contracts.
- One User can receive many Notifications.
- One User can generate many Reports.

7. Conclusion

The database design for the Vendor Reliability Intelligence Platform provides a structured and scalable foundation for managing vendors, procurement, contracts, user authentication, reports, and notifications. The use of relational tables, primary keys, and foreign keys ensures data integrity and supports efficient implementation using PostgreSQL.